Towards a Computational Codicology: A Framework for Manuscript Descriptions

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This poster presents a framework being developed within the ERC-funded **PRIMA project** for systematic codicological descriptions of Early Modern manuscripts: Codicological Model for Early Modern Manuscripts (CoMEMM). While traditional digitisation practices and models like TEI document well textual content, they generally lack the granularity required for detailed codicological research. Our framework complements and extends existing standards to capture comprehensive data on physical and structural features, including writing supports, gathering structures, page preparation, and other key material characteristics.

Framework Overview

Inspired by models like VisColl (Porter et al., 2017; Porter and Campagnolo, 2021), the Ligatus schema for the description of bookbindings (Ligatus, 2007; Velios and Pickwoad, 2012; Velios and Pickwoad, 2020b), and the practices of quantitative and analytical codicology (Andrist et al., 2013; Ornato, 2020), the framework integrates manual and metadata-driven descriptions into computationally analysable datasets. This approach supports both detailed studies of individual manuscripts and comparative analyses across collections. Serialized through an XML schema, the modular structure of the model allows for flexible yet consistent documentation of complex material features, enabling cross-collection analysis and highlighting of production patterns, also integrating external models (such as VisColl full collation models). The framework's design prioritises extensibility, allowing researchers to incorporate new descriptive elements as research methodologies evolve.

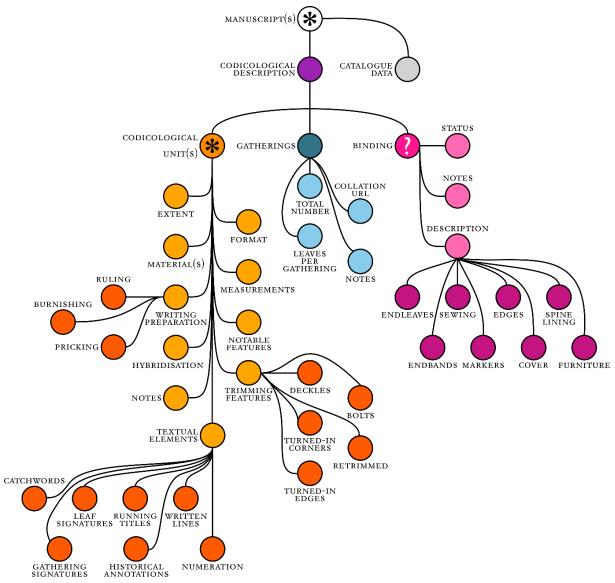

Figure 1. Diagram of the higher-level fields of the CoMEMM framework. Asterisks (*) denote fields that can have one or more instances, while question marks (?) indicate fields with zero or more instances.

At its core, the framework builds on established codicological methodologies and applies stratigraphic principles commonly used in archaeology (Warburton, 2008). It conceptualises the manuscript as a layered artifact, where each element—such as gathering assemblies, ruling patterns, page preparation, or binding structure—contributes to a broader understanding of its construction and use. By capturing the relationships among these features the framework assists researchers in studying production processes and in identifying interventions throughout a manuscript's history. For example, ruling patterns and *mise-en-page* can be systematically analysed to reveal how manuscripts were designed to suit specific purposes, offering insights into scribal practices and manuscript design principles (Ornato, 2020). Additionally, the model facilitates the study of hybridisation and material reuse, repair, and other traces of historical intervention that inform a manuscript's life cycle and evolving role in cultural contexts.

Analytical Capabilities and Applications

The framework integrates elements of quantitative codicology, enabling the systematic analysis of large datasets to uncover macroscopic trends in manuscript production, use, and circulation. This includes identifying correlations between features, such as standardised formats for specific manuscript types or regional variations in techniques like ruling, gathering composition, or layout design. By tracking these attributes across collections, researchers can reveal evolving preferences, production practices, and manuscript functionality over time and space. The structured data also supports the integration of material evidence of use and function, such as staining or annotations, both of which contribute to understanding reader practices and cultural significance of manuscripts beyond their production.

The analytical capabilities extend towards the development of advanced tools for pattern recognition and visualization. For example, like VisColl models, the framework's structured data can be used to create visual representations of page layouts or to enable analyses of ruling systems and identification of specific production practices. These applications provide a basis for linking manuscripts to production centres or regional traditions.

Technical Implementation and Interoperability

The framework is designed to align codicological documentation with computational models, facilitating the integration of disparate datasets. A structured XML schema provides standardised field names and descriptive terminologies for material and structural features, establishing a shared vocabulary (Muzerelle, 2002; Andrist et al., 2013; Velios and Pickwoad, 2020a). By serializing the data in machine-readable formats, the framework supports long-term preservation, data exchange, and computational analysis. This interoperability enhances broader access to and reuse of codicological data and possible integration within TEI documents through namespacing.

Conclusion and Future Implications

By bridging traditional codicological methods with computational approaches, this framework advances digital codicology while fostering interoperability, accessibility, and data integration in cultural heritage research. The resulting datasets support both focused scholarly investigation and broader quantitative analyses, contributing to a deeper understanding of historical book production, usage, and circulation. The framework's extensibility and modular design ensure its adaptability to emerging research methodologies, making it a valuable tool for future investigations into manuscript studies. Furthermore, its potential extends beyond the manuscript book, offering opportunities for interdisciplinary integration with related fields such as book history and cultural heritage studies. As a foundation for evolving methodologies, this framework plays a key role in the growing landscape of digital codicology.

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